

Teo Yu Qi

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WORK

Software Engineer

Jun 2024 – Present

Shopee (Engineering Infrastructure)

Singapore

- Developed and maintained Shopee's infrastructure TechOps platform, serving over 3,000 employees.
- Engineered backend middleware microservice to serve different versions of frontend assets, enabling controlled feature rollouts for canary, grayscale, and A/B testing.
- Optimized frontend performance by injecting Redis-cached runtime configs into HTML templates, eliminating 5 JSON config file requests and improving average framework load time by 25.7%.
- Improved micro-frontend performance by preloading commonly-used bundles using metadata stored in Redis, reducing average framework load time by 5.8%.
- Alleviate engineer pain point by creating a serverless FaaS function to easily toggle release configurations via chat commands on Shopee's internal messaging platform.

Software Engineer Intern

Jan 2023 - Nov 2023

TikTok (Global E-Commerce, Data Intelligence)

Singapore

- Developed and maintained frontend systems for TikTok Shop Creator, Partner, and Affiliate accounts to track and analyze KPI and sales metrics, serving over 500,000 weekly active users with sales.
- Revamped UI of Creator Dashboard's landing page on mobile, contributing to +33.54% increase in user views and +57.66% increase in page views.
- Improved frontend system observability of TikTok Shop Creator, Partner, and Affiliate platforms by adding API request monitoring and alerting.
- Introduced asynchronous data export feature on Partner Compass SEA, with 70 TikTok Shop Partners utilizing the feature and ~650 file exports in the first week of release.

Software Engineer Intern

May 2022 – Jul 2022

DSO National Laboratories

Singapore

- Researched viability of classifying RF pulse using AI/ML algorithms on AMD Xilinx MPSoC.
- Deployed deep-learning model on Avnet UltraZed-EG with Vitis-AI SDK in Python.

OTHERS

Open-Source Contributor, Electron

Nov 2025

- Improved BaseWindow API documentation by clarifying previously undocumented behavior.
- Changes merged into Electron versions 39-x-y, 40-x-y. [PR #48931]

TikTok Tech Immersion 2023

May 2023

- Attended 8-day course on backend, quality assurance and site-reliability engineering.
- Configured HTTP and RPC server in Go as part of course deliverables.

Student Volunteer, Centre for Computing for Social Good & Philanthropy

May 2022 – Jun 2022

- Set weekly quizzes for OTH4199 AI Primer Course. (Beneficiary: Autism Resource Centre - Singapore).

PROGRAMMING LANGUAGES & FRAMEWORKS

Fluent in: TypeScript, JavaScript, Python, React, Node.js, NestJS, Express, Electron, HTML, CSS

Also used: Bash, C, C++, Docker, Go, Java, MySQL, NextJS, PostgreSQL, Redis

EDUCATION

National University of Singapore (NUS)

Aug 2020 – May 2024

Bachelor of Computing in Computer Science (with Honors)

Singapore